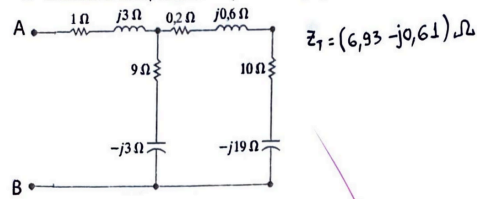


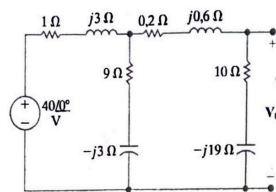
Prova Eletricidade CA - S2

1. Determine a Impedância Equivalente (Z_T) entre A e B.



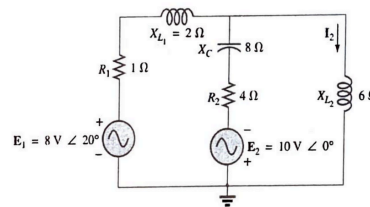
$$Z_T = (6,93 - j0,61) \Omega$$

2. Determine V_0 .



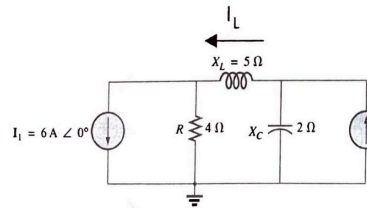
$$V_0 = 40,74 \angle -27,55^\circ \text{ V}$$

3. Qual a corrente I_2 do circuito abaixo:



$$I_2 = 1,27 \angle -86,95^\circ \text{ A}$$

4. Qual o valor de I_L .



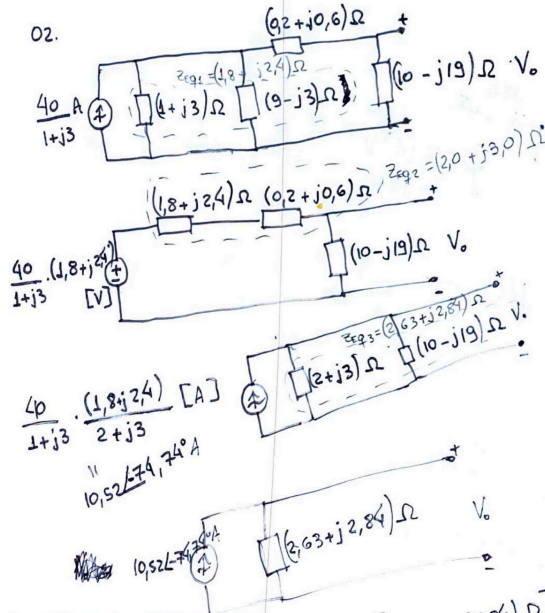
$$I_L = 5,06 \angle -55,30^\circ \text{ A}$$

$$Z_T = (1 \Omega) + (j3 \Omega) + [(9 \Omega - j3 \Omega) \parallel (0,2 \Omega + j0,6 \Omega + 10 \Omega - j19 \Omega)]$$

$$Z_T = (1 + j3) \Omega + [(9 - j3) \Omega \parallel (10,2 - j18,4) \Omega] \quad \checkmark$$

$$Z_T = (1 + j3) \Omega + \frac{1}{\frac{1}{(9 - j3) \Omega} + \frac{1}{(10,2 - j18,4) \Omega}}$$

$$Z_T = (6,93 - j0,61) \Omega \quad \checkmark$$



$$V_0 = (10,52 \angle -74,74^\circ) [(2,63 + j2,84) \Omega]$$

$$V_0 = 40,74 \angle -27,55^\circ \text{ V} \quad \checkmark$$

03.

$$V = \frac{8 \angle 20^\circ}{1+j2} - \frac{10 \angle 0^\circ}{4-j8} \quad V$$

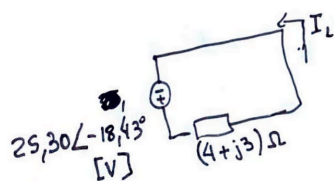
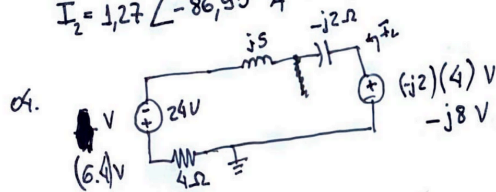
$$\frac{1}{1+j2} + \frac{1}{4-j8} + \frac{1}{j6}$$

$$V = 7,63$$

$$V = 7,64 \angle 3,05^\circ \quad V$$

$$I_2 = V / (j6 \Omega)$$

$$I_2 = 1,27 \angle -86,95^\circ \quad A$$



$$(24-j8)V$$

$$I_L = \frac{205,30 \angle -18,43^\circ}{4+j3} \quad A$$

$$I_L = 5,06 \angle -55,30^\circ \quad A$$